INTERIM FEDERAL SPECIFICATION

BREAKER, FLOURESCENT LAMP

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable, electric-motor driven machines for use in breaking fluorescent lamps.

1.1.1 Federal specification coverage. Federal specifications do not include all types of commodities indicated by the titles of the specifications, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. This specification covers one type of fluorescent tube breaker.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

J-C-580 - Cord, Flexible; and Wire, Fixture, (Electrical, 0-to 600 -Volt Service)
W-C-596 - Connector, Plug, Electrical; Connector, Receptacle, Electrical
CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower)
RR-C-82 - Cans, Ash and Garbage; Corrugated, Taper-Side, Zinc-coated; and covers.
PPP-D-729 - Drums: Metal, 55-gallon (For shipment of Noncorrosive Material).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage (Military Activities)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS.

3.1 Performance. Flourescent lamp breaking machines furnished under this specification shall be capable of safely and continuously breaking all sizes of straight flourescent lamps within the limits of the feeder tubes furnished at a minimum average rate of 120 lamps per hour, depositing the residue in a refuse container. In the process, a means shall be incorporated into the design of the machine to prevent throw-back of glass particles through the feeder tube and to insure that phosphor dust and mercury vapor concentrations released to the surrounding atmosphere do not exceed 0.5 particles per million (p.p.m.) of phosphor dust and 0.1 milligram per cubic meter (mg/cu.m) of mercury vapor, tested as specified in paragraph 4.4.1.

3.2 Construction. Machines shall be comparable in quality and construction to models that have been marketed commercially for at least six months and for which replacement parts are readily available.

3.3 Power data. An electric motor conforming to CC-M-636 shall be securely mounted on the machine, and when used to indirectly drive the breaker mechanism, shall be provided with a means for adjusting belt tension or gear clearance. The motor shall be of a type, class, style, and duty class suitable for the purpose, and unless otherwise specified (see 6.2), shall operate on a 115 volt, 60 cycle, single phase, alternating current. An on-off switch shall be mounted in a readily accessable location on the motor or on the machine.

3.3.1 Power connection. Unless otherwise specified (see 6.2) the motor shall be furnished with a minimum eight-foot-long electrical cord with ground wire conforming to J-C-580, type S. The power line connection end of the electrical cord shall be furnished with a male connector plug, and conforming to W-C-596, Series D.

3.4 Feeder tube. A one-piece or sectional feeder tube of rigid material shall be furnished with each machine. Unless otherwise specified (see 6.2), the feeder tube shall be of sufficient size to receive tubes 2-1/8 inches in diameter, and shall have sufficient length to prevent broken glass from being discharged through the inlet opening. The feeder tube shall be securely mounted to the breaker housing with the inlet end inclined to permit gravity feeding of the lamps into the breaker mechanism. The feeder tube shall be capable of being readily disassembled and reassembled for shipping and storage.

3.5 Breaking component. The blade or impeller used in the machine shall be of adequate material and hardness necessary to resist wear and deformation, tested in accordance with paragraph 4.4.1.

3.6 Refuse container. Machine using liquid as an air pollutant filter medium shall be furnished complete with a liquid tank and perforated disposal basket. The tank and disposal basket shall be made of corrosion resistant material or shall be suitably coated to resist normal environmental corrosion. The base of machines not using liquid as an air pollutant filter medium shall be suitable for use with 32 gallon cans conforming to RR-C-82 and drums conforming to PPP-D-729.

3.7 Weight. The complete machine shall weigh not more than 100 pounds, or shall be furnished with a dolly.

3.8 Dimensions. The height of the inlet end of the feeder tube from the floor, with the machine ready to operate, shall not exceed five feet. The machine shall require a floor area not greater than two feet wide by three feet long with feeder tube(s) removed.

3.9 Guards. All moving parts which are of such nature or so located to be a hazard to operating personnel shall be suitably enclosed or guarded.

3.10 Lubrication. All surfaces requiring lubrication shall be prelubricated prior to acceptance, and except for sealed bearings, shall be provided with a means of being relubricated.

3.11 Marking. Each machine shall be marked conspicuously in a legible and permanent manner with the manufacturer's name and model number of the machine. Component parts not manufactured by the manufacturer of the machine shall bear the identification marking affixed by the manufacturer of the component.

3.12 Replaceable parts. All component parts that are subject to replacement shall be readily accessable and shall be interchangable with the replacement parts stocked by the manufacturer.

3.13 Manuals. A copy of the manufacturer's standard operating and maintenance instructions shall be furnished with each machine.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2 Sampling

4.2.1 Lot. All fluorescent lamp breaking machines of the same model presented for acceptance at one time shall be considered a lot for purposes of sampling.

4.2.2 Sample. One machine shall be selected at random from each lot for inspection and testing to verify compliance with this specification.

4.3 Inspection

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Presence of one or more defects shall be cause for rejection.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any noncomformance with dimensional and weight requirements. Failure to conform to one or more of the requirements shall be cause for rejection.

4.4 Test.

4.4.1 Performance test. Each sample unit shall be tested for performance to determine compliance with this specification in accordance with the following procedures. Failure of the sample to comply with requirements 3.1, 3.3, 3.4 and 3.5 shall be cause for rejection.

4.4.1.1 Equipment required. Necessary equipment to perform the test shall be furnished by the supplier or an acceptable commercial laboratory and shall consist of a clock or watch and instruments capable of collecting and counting dust particles, measuring concentrations of mercury vapor and measuring electrical voltage. The instruments used shall be the products of a reliable manufacturer of instruments and shall be suitable for the tests to be performed. Also required, are 240 fluorescent lamps (used lamps acceptable) of various sizes including a minimum of 25% being the largest size for which the feeder tube was designed to receive, and a refuse container as specified in paragraph 3.6 when the machine does not use liquid as an air pollutant filter medium.

4.4.1.2 Test conditions. Prior to the running of the test, a closed and unventilated room with a maximum ceiling height of ten feet and a cubic foot volume not over 5000 cubic feet will be selected for the test. The actual volume of the room will be calculated and converted to cubic meters by dividing the cubic feet by 35.10 to determine the cubic meter volume. The resulting figure shall be used as a factor in determining mercury vapor concentrations. To insure that phosphor dust and mercury vapor are not present in the test room before the test, samples of the air shall be taken. If necessary, the room will be ventilated, or another test room selected until it is possible to obtain a negative reading. The sample to be tested shall be assembled in operating position in accordance with the manufacturer's instructions and within 8 feet of an electrical outlet designed to receive the male connector plug. The line voltage shall be checked for consistency with the voltage requirements of the motor, and if compatable, the sample shall be connected to the power supply.

4.4.1.3 Test. The test shall consist of noting the starting time, and inserting the 240 lamps supplied for the test into the feeder tube. The lamps shall be inserted into the feeder tube individually and continuously, with no effort made to force the lamps into the breaker mechanism, until all 240 lamps have been broken and the finish time noted. Immediately following, samples of the air in the proximity of the machine shall be taken to determine the machines effectiveness of filtering or absorbing phosphor dust and mercury vapor.

4.5 Examination after test.

4.5.1 Visual examination. A visual examination shall be made of the sample unit after successful completion of the performance test to determine points of undue wear of the components. Any undue wear detected shall be cause for rejection.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified (see 6.2) the breakers shall be prepared for shipment in accordance with the manufacturer's recommendations, which will insure acceptance and safe delivery at the lowest rate to the ultimate destination.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes

6.1 Intended use. Fluorescent tube breakers covered by this specification are intended for use when a sizable quantity of fluorescent lamps are to be disposed of in a safe manner, and compact disposal is desired. However, to insure safe working conditions, it is suggested that the machine be operated in well ventilated areas and for intermittent periods of time.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Motor data, if different (see 3.3).
(c) Power cord and line connection, if different (see 3.3.1)
(d) Feeder tube capacity, if different (see 3.4)
(e) Specific packaging and packing requirements, if different than specified (see 5.1).

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